

JEE MAIN REVISION MASTER: HEIGHT AND DISTANCE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Height And Distance - Core Application	40 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Height And Distance - Core Application (Very High Yield — 40 PYQs)

Core Principle: Key problem pattern in Height And Distance.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due North of the tower is 30° . The height of the tower is $100\sqrt{3}$ m. The distance between the two persons is $100\sqrt{3}$ m." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "From the top A of a vertical wall AB of height 30m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 30° and 60° respectively. The height of the tower PQ is $10\sqrt{3}$ m." [Ans: N/A]
- [2022 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on the ground such that AR is perpendicular to PQ. The angle of elevation of P from R is 30° . The distance AR is $10\sqrt{3}$ m." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due west of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to

- (A) 10
- (B) $(5/2)\sqrt{5}$
- (C) $5\sqrt{5}$
- (D) 5

Correct Answer: None

Solution Analysis: Let's denote the person standing due south as S and the one standing due west as W. Also, let the tower be at point T. From person S's perspective, we have a right triangle $\triangle SPT$. The height of the tower PT is given as 5m, which is the opposite side for angle S. The angle at S is 45° . From the tangent trigonometric ratio, we have : $\tan 45^\circ = (PT / ST) = (5 / ST)$ Since $\tan 45^\circ = 1$, we get $ST = 5$ m. Similarly, from person W's perspective, we have a right triangle $\triangle WPT$. The angle at W is 30° . From the tangent trigonometric ratio, we have: $\tan 30^\circ = (PT / WT) = (5 / WT)$